



Physics-Constrained Thermal Map Prediction for VLSI Chip Design

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Motivation

Thermal hotspots are a **first-order constraint** in chip design: they cause performance throttling, increased leakage current, and reduced device lifetime. Thermal simulation (HotSpot, ANSYS) can take seconds-to-minutes per design-prohibitive inside automated floorplan optimization loops that require $\geq 10^5$ evaluations.

Neural surrogates approximate simulator output in milliseconds, but prior work lacks physics grounding, leaving implausible temperature fields when die geometry or cooling conditions shift outside training.

Our approach: image-to-image regression with a *differentiable physics penalty* that enforces the 2D steady-state heat equation on the predicted thermal map—no simulator calls during training.

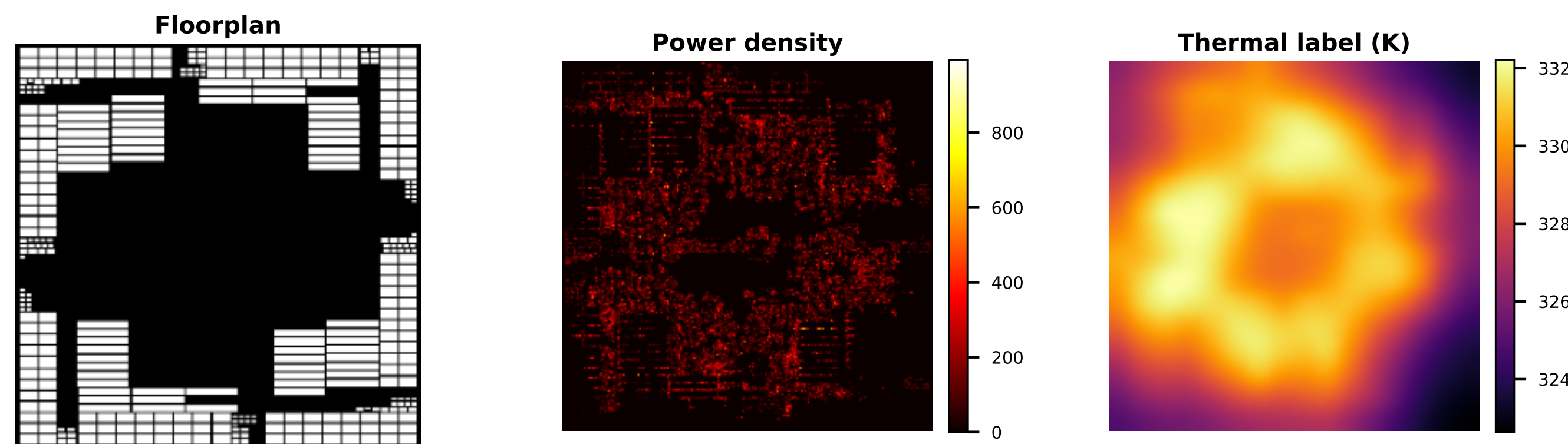


Figure 1. Representative sample (Vortex-large). **Left:** floorplan. **Center:** power-density map. **Right:** HotSpot thermal label (K).

Dataset

Source: CircuitNet 2.0 [1] GPU designs: Vortex-small (96), Vortex-large (61), NVDLA-large (32) - **N=189**.

Labels: HotSpot [3] grid mode (256×256), die thickness 150 μm , $r_{\text{convex}}=0.1$ K/W, power capped at 10 W/tile. Parallelized via Modal serverless workers.

Splits: 80/10/10 train/val/test per family (seed 42) $\Rightarrow$ 153 train, 18 val, 18 test. OOD evaluation uses val+test (36 instances) re-labeled at 4 thicknesses (75, 100, 200, 300 μm) and 4 convective resistances ($r \in \{0.01, 0.05, 0.20, 0.50\}$ K/W).

Normalization: per-family z-score; augmentation via random H/V flips.

Physics-Constrained Loss

The steady-state temperature field satisfies the 2D heat equation:

$$k \nabla^2 T + Q = 0 \quad (1)$$

We approximate the Laplacian on the predicted map $\hat{T}$ via a fixed 3×3 finite-difference convolution:

$$\nabla^2 \hat{T} \approx \mathbf{K} * \hat{T}, \quad \mathbf{K} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (2)$$

The composite training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}}(\hat{T}, T) + \lambda_{\text{phys}} \|k(\mathbf{K} * \hat{T}) + Q\|_F^2 \quad (3)$$

The kernel is a **non-learnable buffer**; gradients flow only through $\hat{T}$. Because $\mathcal{L}_{\text{phys}}$ depends only on the prediction and power input (not the label), it remains active on OOD samples with no extra annotation. We sweep $\lambda_{\text{phys}} \in \{0, 0.01, 0.1, 1.0\}$ across all three architectures — **12 trained models** total.

References

- [1] Zhuomin Chai, Yuxiang Zhao, Wei Liu, Yibo Lin, Runsheng Wang, and Ru Huang. CircuitNet 2.0: An advanced dataset for promoting machine learning innovations in realistic chip design environments. In *ICLR*, 2024.
- [2] Liang Cheng et al. DeepOHeat: Operator learning-based ultra-fast thermal simulation in 3D-IC design. In *DAC*, 2023.
- [3] Wei Huang, Shougata Ghosh, Siva Velusamy, Karthik Sankaranarayanan, Kevin Skadron, and Mircea R. Stan. HotSpot: A compact thermal modeling methodology for early-stage VLSI design. *IEEE Trans. VLSI Systems*, 14(5):501–513, 2006.
- [4] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks. *Journal of Computational Physics*, 378:686–707, 2019.
- [5] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015.
- [6] Wenzhe Shi et al. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. In *CVPR*, 2016.

Model Architectures

All models take a 2-channel 256×256 floorplan+power image as input and produce a 1-channel thermal heatmap. We evaluate three architectures to isolate the contributions of spatial hierarchy and skip connections.

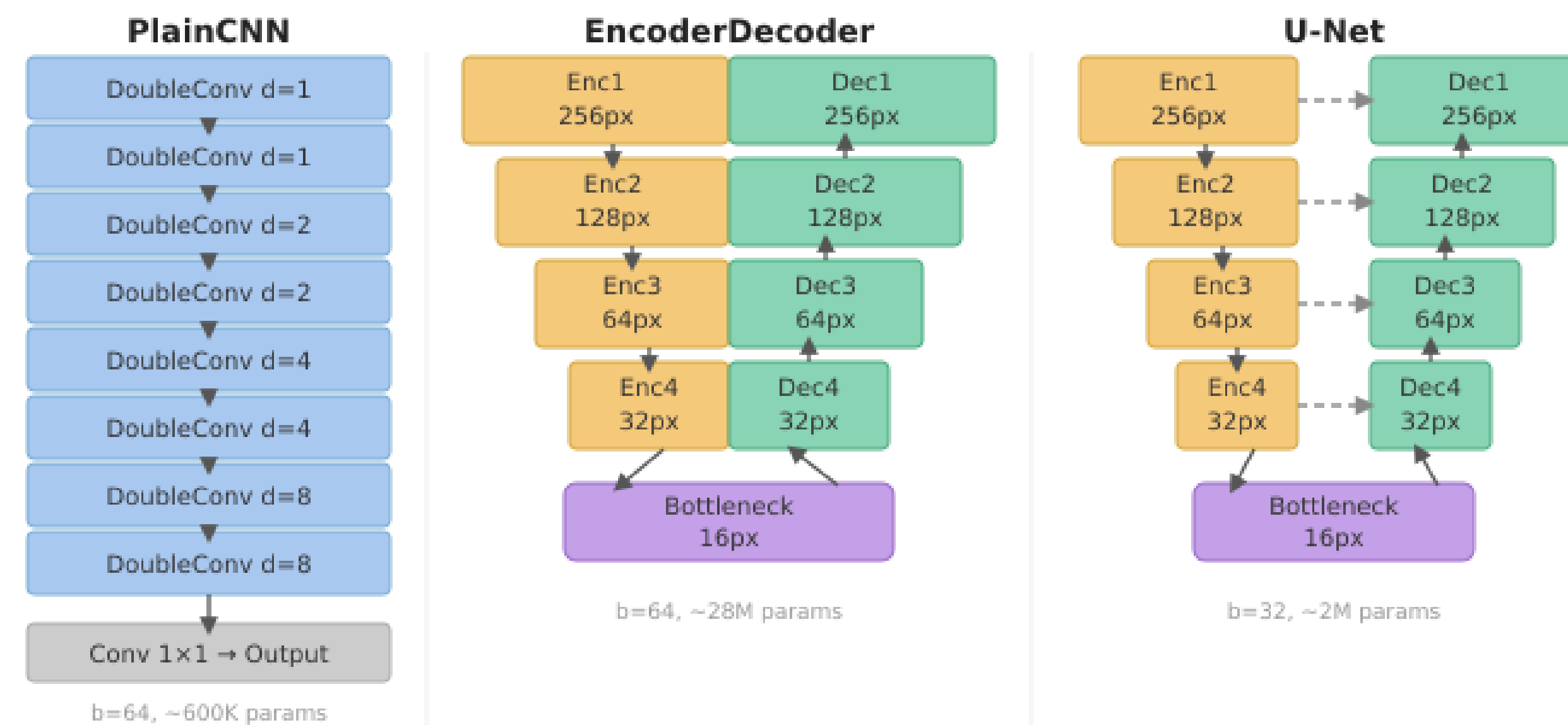


Figure 2. The three architectures. Encoder blocks (orange) use MaxPool2d; decoder blocks (green) use PixelShuffleUp [6]. The key structural difference is the presence (U-Net) or absence (EncoderDecoder) of lateral skip connections.

- **PlainCNN** ($b=64$, $\approx 600\text{K}$ params): eight DoubleConv blocks at full resolution, dilations [1, 1, 2, 2, 4, 4, 8, 8]; no downsampling or skip connections.
- **EncoderDecoder** ($b=64$, $\approx 28\text{M}$ params): four-stage encoder PixelShuffleUp decoder; no skip connections.
- **U-Net** ($b=32$, $\approx 2\text{M}$ params): identical encoder/decoder + skip connections; bottleneck **Dropout2d** ($p=0.3$).

In-Distribution Results

Model	RMSE (K) ↓	SSIM ↑	Hotspot IoU ↑
PlainCNN	2.420	0.987	0.630
EncoderDecoder	1.213	0.999	0.781
U-Net ($\lambda=0$, MSE-only)	1.008	0.995	0.819
U-Net ($\lambda=0.01$)	0.895	0.996	0.841

Table 1. Test-set results. RMSE in Kelvin (after denormalization). Bold = best.

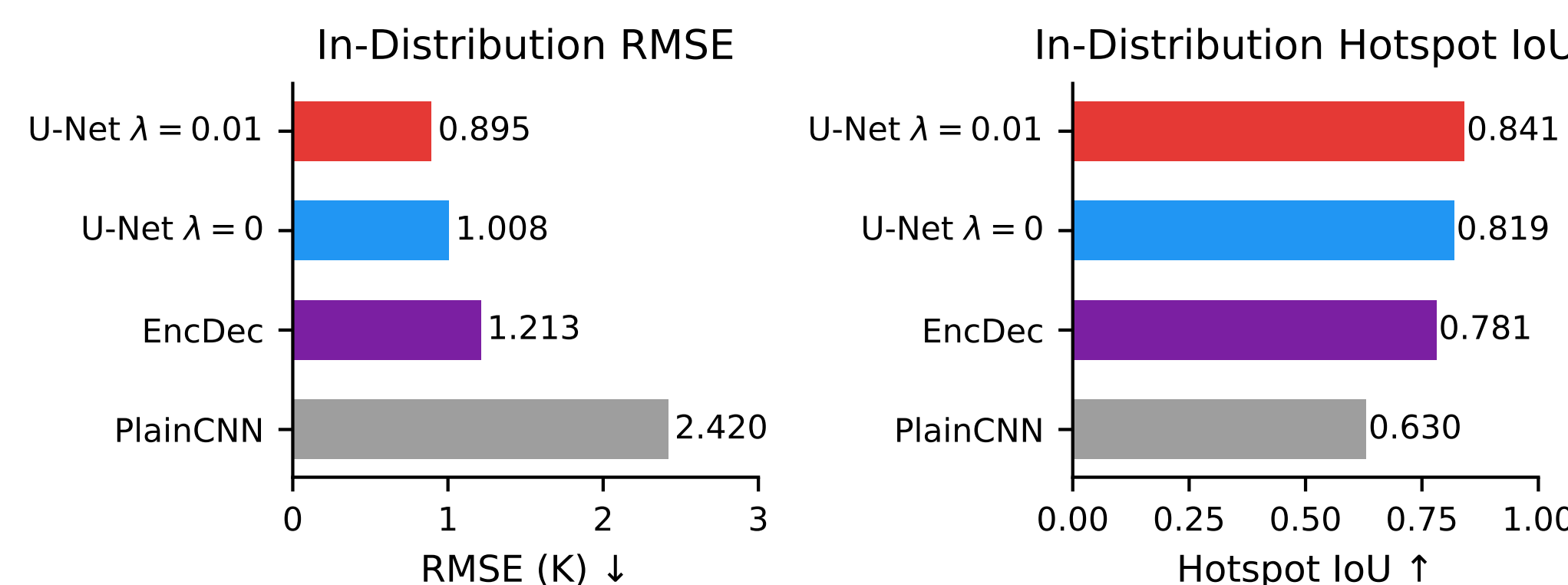


Figure 3. In-distribution RMSE and hotspot IoU across all four model variants. U-Net $\lambda=0.01$ achieves the best score on both metrics.

Key findings:

- Architecture dominates: U-Net/EncDec are 2–2.5× better RMSE than PlainCNN regardless of λ .
- Skip connections improve U-Net over EncDec by **17% RMSE** and **3.8 pp hotspot IoU**.
- Physics penalty at $\lambda=0.01$ reduces U-Net RMSE by **11.2%** (1.008 $\rightarrow$ 0.895 K) and hotspot IoU **+2.2 pp**.

Out-of-Distribution Generalization

OOD behavior is **asymmetric** across the two parameter axes:

Model	Convection resistance (K/W)				Die thickness (μm)			
	$r=0.01$	$r=0.05$	$r=0.20$	$r=0.50$	75	100	200	300
U-Net $\lambda=0$	1.595	1.259	1.203	3.847	6.592	4.299	3.137	9.376
U-Net $\lambda=0.01$	1.458	1.124	1.200	3.926	6.428	4.136	3.252	9.504

Table 2. OOD RMSE (K). Bold = better. In-dist baseline: $\lambda=0$: 1.008 K, $\lambda=0.01$: 0.895 K.

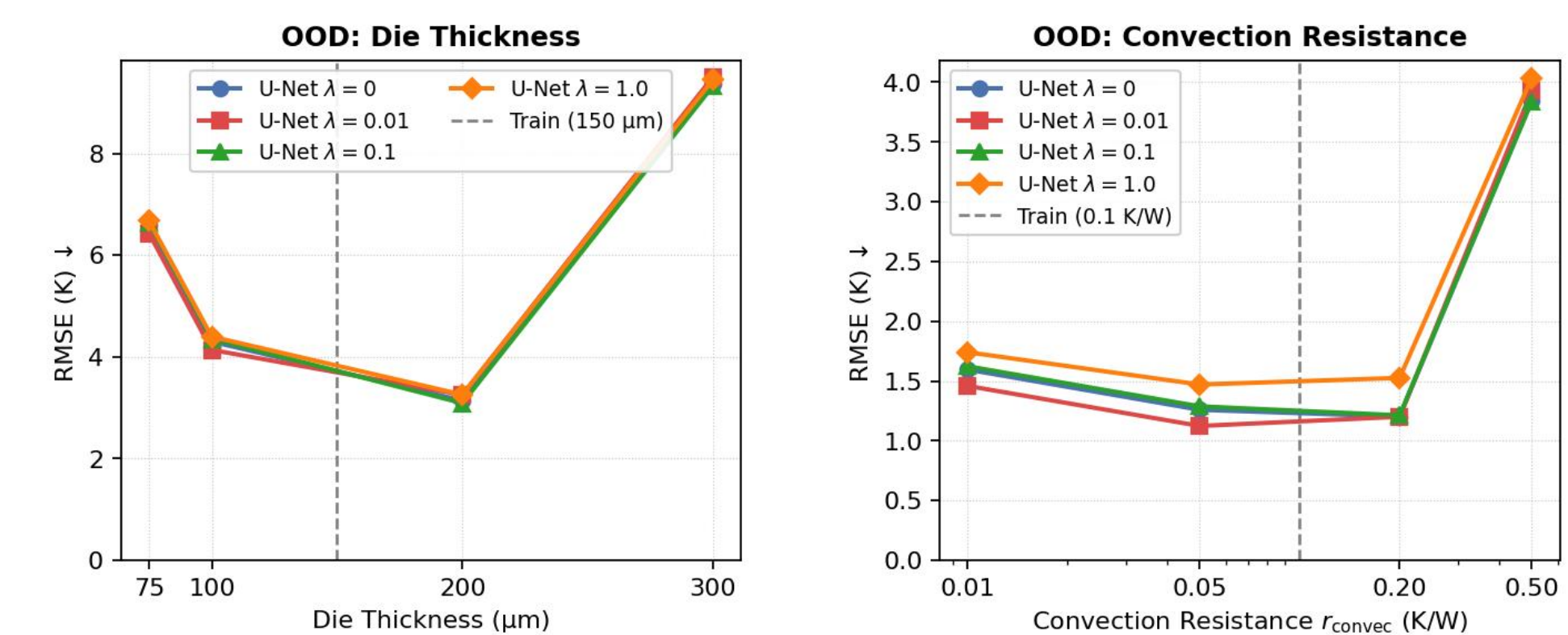


Figure 4. **Left:** thickness OOD - all variants degrade identically. **Right:** convection-resistance OOD - $\lambda=0.01$ outperforms at moderate shifts (0.01–0.05 K/W).

Ablation & Qualitative Results

Metric	U-Net			EncDec	
	$\lambda=0$	$\lambda=0.01$	$\lambda=0.1$	$\lambda=1.0$	$\lambda=0.1$
RMSE (K) ↓	1.008	0.895	1.036	1.303	1.190
IoU ↑	0.819	0.841	0.817	0.795	0.842

Table 3. $\lambda=0.01$ minimizes RMSE. Over-weighting physics ($\lambda=1.0$) degrades accuracy below the MSE-only baseline.

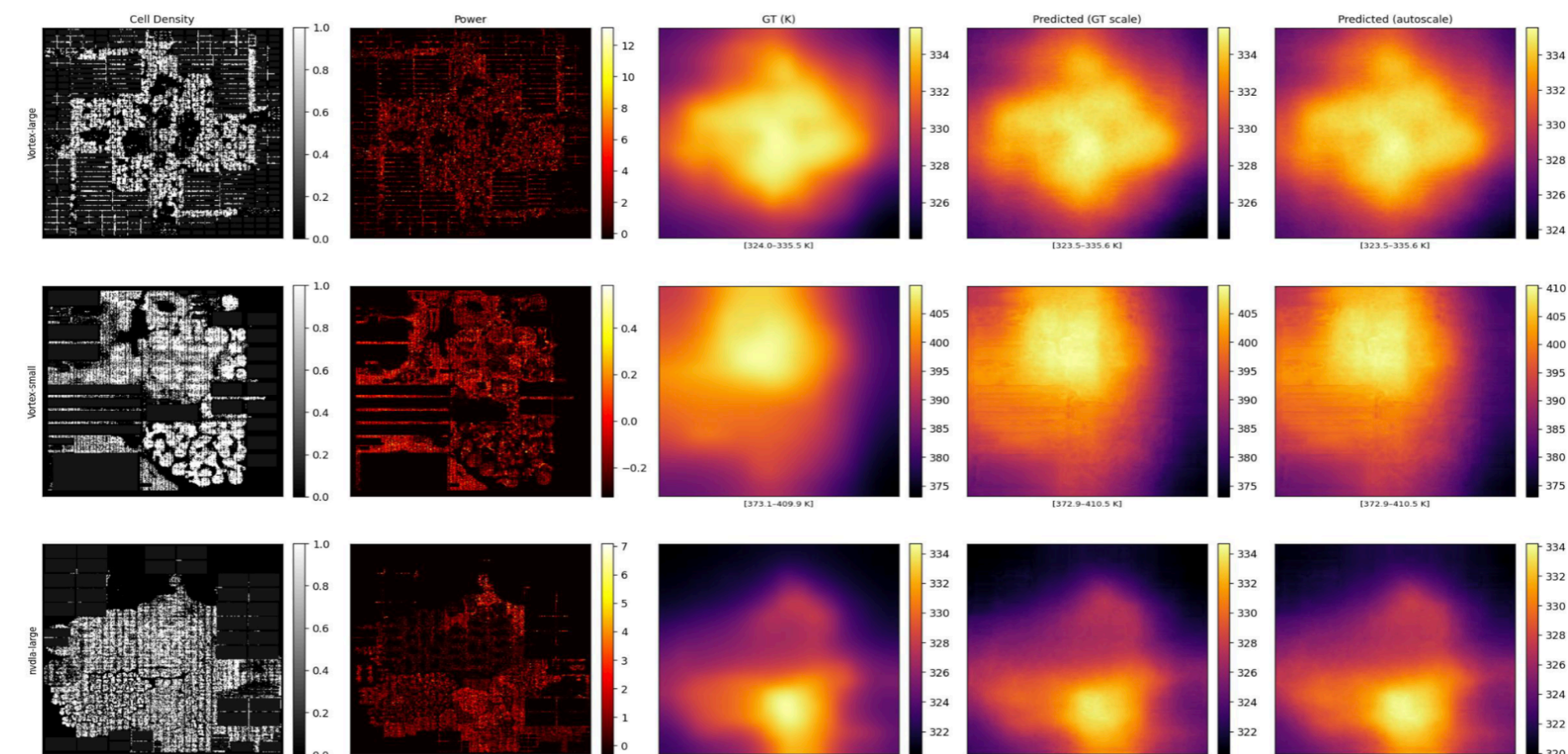


Figure 5. U-Net ($\lambda=0.01$) predictions per chip family. Rows (top to bottom): Vortex-small, Vortex-large, NVDLA-large. Columns (left to right): floorplan, power density, GT thermal (K), prediction (GT scale), prediction (autoscale).

Conclusions

- **U-Net + physics loss** ($\lambda=0.01$) achieves **0.895 K RMSE** and **0.841 hotspot IoU** - 11% RMSE improvement over MSE-only.
- Skip connections and physics penalty contribute **orthogonally**: skips improve RMSE; Laplacian penalty improves hotspot IoU.
- OOD benefit is **asymmetric**: 8–11% gain for moderate HTC shifts; *no* gain for die-thickness extrapolation.
- Robust geometry generalization requires stronger physics priors or geometry-aware architectures.